

FREDDY: Fast Reduction of Elements for Data-driven Yielding

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Abstract

The current era of big data and machine learning, especially deep learning models, has driven unprecedented advances in various industries, from population genomics to advertising. Deep learning models have been widely used in a variety of fields, such as machine learning and genetics. However, it is difficult to train a machine learning model on a large scale, due to the complexity of the datasets. In this paper, we present a categorization of coreset selection methods, divided into three paradigms: (I) *training-free methods*, (II) *training-oriented methods*, and (III) blind selection. The main contribution of this paper is to present an overview of the current state of the art in the field of deep learning, with a focus on the use of a subset selection method for large-scale datasets, with the aim of reducing training time and resource requirements.

1 Introduction

The combination of big data and machine learning (ML), especially deep learning, has driven unprecedented advances in many domains, from population genomics to advertising Huggins et al. (2016); Weinreich et al. (2025). However, the volume of data used to train machine learning models can be prohibitively expensive, both in terms of computational cost, energy consumption and carbon emissions associated with the extensive use of GPUs Moser et al. (2025); Huggins et al. (2016).

The training process consists of guiding the model on its journey through parameter space toward regions that enhance predictive performance. This process can be expensive when training machine learning (ML). A classical solution to the problem under consideration relies on stochastic gradient descent (SGD) algorithm. The purpose of the SGD is to accelerate the optimization problem by estimating the full gradient via mini-batches of training data.

In order to speed up ML training, the literature presents some strategies such as distributed training. Distributed training requires decisions, such as defining weight and data partitioning strategies, as well as gradient update policies. Communication between computation nodes emerges as a significant bottleneck. The necessity of constant communication between computing devices to maintain cohesion during backpropagation has the potential to reduce the efficiency of the training process Xing et al. (2015); Aaron et al. (2018).

An alternative way to overcome data volume challenge is to build ML models over representative samples – data summarization. The idea is to provide a subset from the original dataset. Recent results in machine learning model training indicate that large volumes of data do not necessarily produce better models, as performance can be compromised by irrelevant examples, noise, and redundancies Weinreich et al. (2025).

Reducing the volume of data used in the training helps the model complete epochs faster, freeing up resources, and accelerating the cycle. However, this time saving may come at the cost of lowering the final performance. The central question is whether it is feasible to reduce the volume of data without reducing the quality of the output. Data summarization techniques as active learning and coreset selection help to optimize the machine learning model training process by reducing the amount of data needed. Active Learning tackles this issue by employing an iterative cycle in which the model itself identifies and actively requests more valuable instances in order to reduce data redundancy. On the other hand, coreset selection algorithms aim to find a representative subset of data such that certain properties of the original dataset are preserved, Huggins et al. (2016); Chai et al. (2023); Wang et al. (2025); Mirzasoleiman et al. (2020).

Despite coreset selection literature that presents evidences of success on saving training time, its implementation often encounters significant obstacles, which have the potential to compromise the

effectiveness and feasibility of the method. Computational cost is a major obstacle, which can turn to be prohibitive for many approaches, particularly those based on gradient approximation. The preprocessing stage of coreset computation is computationally intensive, as it involves calculating pairwise distances between each item pair in the dataset. For applications that depends on a substantial information volume, this overhead has the potential to result in the sampling process lasting more than the training process itself.

In this paper, we present Freddy, a novel coreset selection algorithm based on gradient approximation. The objective of Freddy is to enhance selection efficiency through gradient approximation under resource constraints. Our main contribution is to perform the training gradient estimation within mini-batches. This mechanism was designed to reduce computational requirements and execution time, making coreset selection more feasible for large-scale applications and aligning with the need for more sustainable machine learning models.

Experimentally, Freddy was able to outperform state-of-the-art methods in both time and quality of the models produced. The algorithm was tested on both regression and classification problems. In some scenarios, Freddy was able to increase the efficiency of colorset selection, achieving more than $21\times$ speed up compared to CRAIG Mirzasoleiman et al. (2020).

This paper is organized as follows: Section 2 positions FREDDY as a variation of CRAIG Mirzasoleiman et al. (2020), focused on gradient approximation, but with improvements in temporal efficiency through a dataset partitioning strategy and gradient estimation via mini-batches. Section ?? presents the theoretical framework for FREDDY. It defines the objective function for selecting the minimum subset that best approximates the gradient of the training dataset and argues that a greedy heuristic can provide an approximation to this combinatorial optimization problem. Section ?? describes the coreset selection process performed by FREDDY, detailing the partitioning strategy as a greedy two-loop procedure for coreset selection based on mini-batches, with the aim of speeding up the selection. Finally, Section ?? presents the experimental methodology used to evaluate the efficiency of FREDDY in classification and regression tasks.

2 Related Work

The present study is situated within the broader context of dataset summarisation for machine learning training. To elaborate further, the specific research interest is focused on coreset algorithms and their impact on the efficiency of machine learning model training. This section intends to provide a concise overview of existing definition and categorisation of coreset methods. Additionally, it aims to place FREDDY within the broader context of coreset selection for machine learning literature.

The objective of summarising training datasets is to identify a small and representative subset of a larger dataset, with the aim of preserving the structures or distributions of the original data. The subset is capable of preserving the generalization ability of the trained model. The literature on coreset selection frequently presents two primary categories of method: (I) Active Learning and (II) coreset selection. According to Weinreich et al. (2025), active learning is concerned with identifying which unlabelled samples are most significant to the training process and, and therefore should be assigned a label. On the other hand, coreset selection aims to create a subset S of a dataset T (where $|S| \ll |T|$), such that a model trained on S generalizes as well as a model trained on the complete dataset T Huggins et al. (2016).

According to Moser et al. (2025), research on coresets has evolved significantly since its origins in computational geometry, where the problem was initially proposed to calculate the smallest coverage for a set of points. The objective was to identify the smallest subset of points required to approximate a given surface. In the domain of machine learning, the loss function can be conceptualised as a surface. In this sense, a coreset is defined as a subset that enables the approximation of the original surface described by a given loss function. Literature on coreset selection commonly presents solutions addressing specific tasks, such as *clustering* Har-Peled and Kushal (2005), classification Huang et al. (2019), and tabular data summarization Wang et al. (2022). Algorithms such as k-Center Greedy Har-Peled and Mazumdar (2004) aim to cover the data space with a minimum subset of points, prioritizing geometric coverage.

Initially in this section, we explore the taxonomies proposed by Moser et al. (2025) and Weinreich et al. (2025). These taxonomies categorize methods based on their properties and selection characteristics. Next, we will introduce some techniques, classified by Weinreich et al., such as error-

based selection, information-theoretic selection, similarity-based selection, distribution-based selection, confidence-based selection, and learning-based selection. Finally, we will position FREDDY within this context, comparing its characteristics with the other approaches presented and highlighting its contribution to the efficiency of coreset selection.

2.1 Taxonomy and classification

A taxonomy is proposed in Moser et al. (2025) that divides coreset selection methods into three broad paradigms. The initial classification is designated as training-free methods, which employ the intrinsic properties of the dataset, such as distribution or geometry, to select the subset of data. The second category is referred to as training-oriented methods, which rely on information generated by a trained auxiliary model, such as loss function values or gradient norms, to assign a relevance score to each element of the dataset. The third is blind selection. This approach complements training-oriented methods selecting subsets without the need for previously labelled data. In many cases, pseudo-labels are employed to estimate the importance of the sample.

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Concurrently, Weinreich et al. (2025) present a categorisation of training instance selection methods based on the dependence of the trained model’s inference – *a priori* and *a posteriori* methods. *A priori* methods select samples prior to model inference and can be data-oriented (where quality computation does not require model information) or model-oriented (using model information but without complete inference). *A posteriori* methods, perform selection after model inference, thereby creating a dependency relationship with the model prediction to decide whether new sample should be selected.

As Weinreich et al. (2025), coreset methods can also be classified according to their properties and applications. The authors proposes coreset methods classification outlined as follows:

- **Error-Based Selection:** Characterizes data quality by the difference between the trained model’s prediction and the original labels. This set of methods usually has low computational complexity, but its effectiveness depends on the proper threshold definition, being more applicable in simple linear systems.
- **Confidence-Based Selection:** This methods are more suitable on situations where the original label is not known. The data is selected based on measures of the model’s uncertainty or model’s confidence.
- **Information-Theoretic Selection:** Samples are selected by the amount of information they contain about a specific model parameter. It requires significant prior knowledge of the system for effective implementation, and therefore are more difficult to implement.
- **Similarity-Based Selection:** Evaluates each sample based on its similarity to other data points, using distance or correlation metrics. This approach can be used to promote a more diverse coreset. Can also be applied into outlier detection, or deduce dataset’s noise.
- **Distribution-Based Selection:** Defines the selection strategy based on the distribution characteristics of the entire dataset, seeking a more comprehensive view of the underlying distribution of the data. It generally aims to increase the robustness of the estimate against outliers and measurement noise.
- **Learning-Based Selection:** Uses information from auxiliary models to learn the data selection strategy adaptively. This approach is highly flexible for different applications.

Table 1 systematizes the categories proposed in both surveys cited, focusing on the methods and their main hypotheses.

Table 1: Categories and Hypotheses of Data Selection and Coreset Methods

Method	Category (Moser et al. (2025))	Classification (Weinreich et al. (2025))	Main Hypothesis / Key Idea
Herding	Training-Free	Data-Driven (A Priori)	Selects samples sequentially to preserve statistical properties, minimizing the discrepancy between means Welling (2009).
k-Center Greedy	Training-Free	Data-Driven (A Priori)	Finds maximum coverage under geometric measure; suitable for clustering Har-Peled and Mazumdar (2004).
Forgetting	Training-Oriented	Model-Oriented (A Posteriori)	Counts “forgetting” events and selects samples with the highest counts Moser et al. (2025).
GraNd	Training-Oriented	Model-Oriented (A Posteriori)	Selects samples with maximum contribution to the loss function during training, based on gradient norms Moser et al. (2025); Paul et al. (2021).
DeepFool	Training-Oriented	Model-Oriented (A Posteriori)	Prioritizes data close to the decision boundary using adversarial perturbations Moosavi-Dezfooli et al. (2016).
CRAIG	Training-Oriented	Model-Oriented (A Posteriori)	Selects a subset of data whose weighted gradients approximate the total gradient of the dataset Mirzasoleiman et al. (2020).
GradMatch	Training-Oriented	Model-Oriented (A Posteriori)	Directly minimizes a gradient matching error Killamsetty et al. (2021a).
RETRIEVE	Training-Oriented	Model-Oriented (A Posteriori)	Uses two-level optimization to select the most informative unlabeled samples, aiming for better generalization Killamsetty et al. (2021b).
ELFS	Blind Coreset Selection	Model-Oriented (A Priori)	Uses pseudo-labeling via deep clustering to estimate importance scores without true labels Moser et al. (2025).

We propose FREDDY, a novel coreset selection algorithm that aim to select representative subsets that approximate the full gradient – similarity-based. It implement a variation of CRAIG Mirzasoleiman et al. (2020), designed to address its key limitations on time consuming. While CRAIG has demonstrated effectiveness in reducing training time, it remains constrained by computational overhead and scalability issues due to it preprocessing needs. The gradient approximation presented by CRAIG relies on pairwise distance calculations bettween feature vectors in order to provide the the method de information necessary to the gradient estimation. It is noteworthy that this calculation can restrict time efficiency and the algorithm’s scalability.

FREDDY introduces a dataset partitioning strategy that enables the gradient estimation over a small sample from the original dataset. This change enables to reduce coreset selection time consuming while preserves the accuracy reported on CRAIG’s original work Mirzasoleiman et al. (2020).

3 Problem Formulation

Let $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$ denote a dataset of n data objects, where each object is represented exclusively by a d -dimensional feature vector. We consider the fully unsupervised setting: no labels, loss values, or trained models are assumed to be available. The objective is to select a subset $S \subseteq X$ with cardinality at most K that provides a compact yet representative summary of the original dataset.

We formulate data selection as the maximization of a set function

$$F(S) = F_{\text{FL}}(S) + \lambda F_{\text{ICEL}}(S), \quad (1)$$

where F_{FL} encourages global representativeness, F_{ICEL} promotes geometric diversity, and $\lambda \geq 0$ controls the trade-off between these two criteria.

Facility Location Term. The first component, F_{FL} , is a facility location function defined as

$$F_{\text{FL}}(S) = \sum_{i=1}^n \max_{j \in S} s(x_i, x_j). \quad (2)$$

For each data point $x_i \in X$, this term computes its similarity to the most similar selected representative $x_j \in S$. The similarity function $s(\cdot, \cdot)$ is derived from a distance metric (e.g., cosine similarity) and takes values in $[0, 1]$. Intuitively, Eq. (2) measures how well the selected subset S covers the entire dataset: every point contributes according to how close it is to its nearest representative in S .

Intrinsic Coverage Estimation Loss (ICEL). While facility location favors representativeness, it does not explicitly encourage the selection of geometrically informative or complex regions of the feature space. To address this, we introduce a modular regularization term,

$$F_{\text{ICEL}}(S) = \sum_{j \in S} u_j, \quad (3)$$

where u_j is a scalar score associated with element x_j . In this work, u_j is defined as the expected distance between x_j and other data points, computed purely from feature geometry. Points located in sparse or high-variance regions tend to have larger u_j values and are therefore favored by this term. Importantly, ICEL is independent of the selected set S and depends only on the geometry of X .

The objective function in Eq. (1) is monotone and submodular. However, computing exact marginal gains requires evaluating similarities between each candidate and all n data points, which is computationally infeasible for large datasets. In the next section, we introduce a stochastic optimization strategy that overcomes this limitation.

Perfeito. Abaixo está a Seção 4 completamente reescrita, com o nível de didatismo e explicitação esperado em VLDB, alinhada à formalização, sem vaziar detalhes de implementação, e incluindo um algoritmo em pseudo-código claro e autocontido.

O texto explica o papel de cada etapa, o significado das variáveis, e por que cada aproximação é feita, deixando o código como mero detalhe posterior.

Section 4 — FREDDY: Fast Reduction of Elements for Data-driven Yielding

4 FREDDY: Fast Reduction of Elements for Data-driven Yielding

This section presents FREDDY, a stochastic greedy algorithm for scalable data selection that maximizes the objective defined in Eq. (1). FREDDY is designed to operate exclusively on feature vectors and avoids computing or storing full similarity matrices. Its central principle is to approximate marginal gains and coverage updates using random mini-batches, while preserving the geometric interpretation of facility location.

4.1 Stochastic Greedy Optimization

Let S_t denote the subset selected after t iterations. At the core of the optimization problem lies the facility location coverage state, defined for each data point $x_i \in X$ as

$$m_i(S_t) = \max_{j \in S_t} s(x_i, x_j), \quad (4)$$

which measures how well x_i is represented by the current selection. Intuitively, $m_i(S_t)$ stores the similarity between x_i and its closest representative in S_t .

Maintaining this state exactly for all n data points is computationally prohibitive at scale. FREDDY addresses this challenge by maintaining $m_i(S_t)$ as a *stochastic state*, updated only for data points observed in randomly sampled mini-batches. This approximation is unbiased in expectation and enables scalable optimization.

Mini-batch Sampling. At each iteration t , FREDDY samples a mini-batch $B_t \subset X$ uniformly at random, with $|B_t| \ll n$. Mini-batches serve two purposes: (i) estimating marginal gains and (ii) updating the coverage state. Crucially, mini-batches are not used to restrict the domain of the objective function; they are used solely as Monte Carlo estimators.

Marginal Gain Estimation. For a candidate element $e \notin S_t$, the marginal contribution of the facility location term is estimated using the mini-batch B_t as

$$\hat{\Delta}_{\text{FL}}(e \mid S_t) = \frac{n}{|B_t|} \sum_{x_i \in B_t} \max(0, s(x_i, x_e) - m_i(S_t)). \quad (5)$$

The term inside the summation measures the improvement in coverage of x_i that would be obtained by adding x_e to the selected set. The scaling factor $\frac{n}{|B_t|}$ ensures that the estimate is unbiased with respect to the full dataset.

The total estimated marginal gain additionally incorporates the modular ICEL term:

$$\hat{\Delta}(e \mid S_t) = \hat{\Delta}_{\text{FL}}(e \mid S_t) + \lambda u_e, \quad (6)$$

where u_e is the intrinsic geometric score of x_e .

Stochastic Coverage Update. Once an element e is selected, FREDDY updates the coverage state only for points in the current mini-batch:

$$m_i(S_{t+1}) \leftarrow \max(m_i(S_t), s(x_i, x_e)), \quad \forall x_i \in B_t. \quad (7)$$

This update corresponds to a Monte Carlo approximation of the exact facility location update. Across iterations, every data point is updated in expectation, resulting in a consistent approximation of the true coverage state.

Lazy Greedy Selection. To reduce the number of marginal gain evaluations within each mini-batch, FREDDY employs a lazy greedy strategy. Candidates are maintained in a priority queue ordered by their most recent estimated gains. An element is re-evaluated only when it reaches the top of the queue, significantly reducing computation while preserving greedy behavior in expectation.

4.2 Algorithm Description

Algorithm 1 summarizes the FREDDY procedure. The pseudo-code emphasizes the logical structure of the method and omits implementation-specific details such as data structures and vectorized operations.

FREDDY can be viewed as a stochastic submodular maximization algorithm with a noisy oracle and a stochastic auxiliary state. By decoupling the estimation of marginal gains from the maintenance of the coverage state, the algorithm achieves scalability while preserving the geometric intuition of facility location. Importantly, FREDDY relies solely on feature geometry and does not require labels, model training, or clustering, making it suitable for large-scale data management and learning pipelines.

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Algorithm 1: FREDDY: Stochastic Greedy Data Selection

Input: Dataset $X = \{x_1, \dots, x_n\}$, budget K , batch size b , trade-off parameter λ

Output: Selected subset S

Initialize $S \leftarrow \emptyset$;

Initialize coverage state $m_i \leftarrow 0$ for all $x_i \in X$;

while $|S| < K$ **do**

 Sample a mini-batch $B \subset X$ of size b ;

 Initialize a priority queue Q with elements in B ;

while Q not empty and $|S| < K$ **do**

 Extract candidate e with highest stored priority from Q ;

 Estimate marginal gain $\hat{\Delta}(e | S)$ using B ;

if e has highest estimated gain **then**

$S \leftarrow S \cup \{e\}$;

 Update $m_i \leftarrow \max(m_i, s(x_i, x_e))$ for all $x_i \in B$;

else

 Update the priority of e in Q ;

return S ;

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